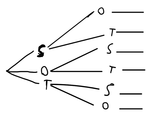
①

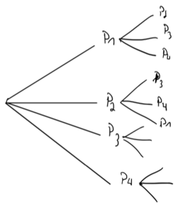



$3 \cdot 2 \cdot 1 = 6$

---

②

$4 \cdot 3 = 12$

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③ $E_1\ E_2$ $\quad E_2\ E_3$ $\quad E_3\ E_4$

$E_1\ E_3$ $\quad E_2\ E_4$

$E_1\ E_4$

6

---

Pg 14/46 :

ex 1) Arrangement avec répétition

$D_{26}^2 \cdot D_{10}^5 = 26^2 \cdot 10^5 = 67\ 600\ 000$

2) Permutation sans répétition

$P_{11} = 11! = 39\ 916\ 800$ mots

3) Combinaison sans rép

$C_{45}^6 = \frac{45!}{39! \cdot 6!} =$

4) Arrangement sans rép

$A_{15}^3 = \frac{15!}{12!} = 15 \cdot 14 \cdot 13$

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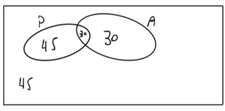


Pg 19, je pense

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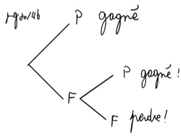
Pg 20/46

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Pg 21/46

| | RC | S | totaux |
|---|---|---|---|
| I | 0,35 · 76 = 57 | 19 | 76 |
| E | 33 | 3 | 36 |
| totaux | 90 | 22 | 112 |

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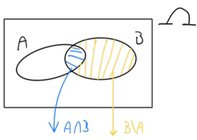


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Pg 29/46

① $A = \{(1;1);(1;2);(2;1);(3;1);(1;3);(2;2)\}$

$P(A) = \frac{\#A}{\#\Omega} = \frac{6}{36} = \frac{1}{6}$

② $\Omega = \{P;F\}$

$P(F) = 3 \cdot P(P)$

$P(\Omega) = 1 \iff P(F) + P(P) = 1$

$\iff 3 \cdot P(P) + P(P) = 1$

$\iff P(F) = \frac{3}{4}$

---

③ $C = C_1 \cup C_2 \cup \dots C_{50}$

où $C_i$ = "faire i fois un 6"

$P(C) = P(C_1) + P(C_2) + P(C_{50}) \rightarrow$

$C^c$ = "faire aucun 6"

$P(C) = 1 - P(C^c)$

$= 1 - \left(\frac{5}{6}\right)^{50}$

$= 0,9989$

---

④

$P(A) = \frac{\text{nbre cas favorables}}{\text{nbre cas possibles}} = \frac{C_{13}^3}{C_{52}^3} = \frac{\frac{13!}{10!\,3!}}{\frac{52!}{49!\,3!}} = \frac{13 \cdot 12 \cdot 11}{52 \cdot 51 \cdot 50} = 0,0129$

$P(B) = \frac{C_{13}^1 \cdot C_{13}^2}{C_{52}^3} = \frac{13 \cdot \frac{13!}{11!\,2!}}{\frac{52 \cdot 51 \cdot 50}{3!}} = \frac{13 \cdot 13 \cdot 6}{52 \cdot 17 \cdot 25} = 0,0459$

C 2 cas :

$C_1$ : "la D ♢ a été tirée"

$C_2$ : "la D ♢ n'a pas été tirée"

$P(C_1) = \frac{C_1^1 \cdot C_1^1 \cdot C_{50}^1}{C_{52}^3} = \frac{1 \cdot 50}{C_{52}^3}$

$P(C_2) = \frac{C_1^1 \cdot C_{12}^1 \cdot C_3^1}{C_{52}^3} = \frac{1 \cdot 12 \cdot 3}{C_{52}^3} = \frac{36}{C_{52}^3}$

$P(C) = P(C_1 \cup C_2) \rightarrow$ blis points

$= P(C_1) + P(C_2) = \frac{86}{C_{52}^3} = 0,0039$

$P(D) = \frac{C_4^3 \cdot C_{13}^1 \cdot C_{13}^1 \cdot C_{13}^1}{C_{52}^3} = \frac{4 \cdot 13^3}{C_{52}^3} = 0,397$